

Question			Marking details]	Marks available				Maths	Prac
				AO1	AO2	AO3	Total		
5	(a)		Work / charge or energy / charge or energy / coulomb (1) Don't accept joules / coulomb Context i.e. energy transferred [or converted] from electrical [potential] / work done..... [between the points] not a freestanding mark (1)	2			2		
	(b)	(i)	1.62 [V] or emf	1			1		
		(ii)	$I = \frac{1.38}{1.50} [=0.92 \text{ A}]$ or by implication if answer correct to 2 sf (1) $V_r = 1.62 \text{ V} - 1.38 \text{ V} = [0.24 \text{ V}]$ or $R + r = \frac{1.62 \text{ V}}{0.92 \text{ A}} = [1.76 \Omega](1)$ $r = 0.26 [\Omega]$ (1)		3		3	3	
		(iii)	$t = \frac{750 \text{ J}}{EI}$ (1) or by implication ecf $t = 500 \text{ s}$ [503 s] (1) [590 s indicates wrong pd used so award 1 mark only]		2		2	2	
		(iv)	Total circuit resistance = 0.26Ω ecf + 0.75Ω or by implication (1) $V_R = 1.20 \text{ [V]}$ (1) or $0.3 \Omega + 0.75 \Omega$ gives $V_R = 1.16 \text{ [V]}$		2		2	2	
	(c)	(i)	$I = \frac{6.0 \text{ V}}{200 \Omega + 850 \Omega} [= 5.71 \text{ mA}]$ or $V_{200} = \frac{200}{200+850} 6.0 \text{ V}$ (1) $V_{200} = 1.14 \text{ [V]}$ (1) $V_{\text{LDR}} = 4.86 \text{ [V]}$ but V_{200} not given award 1 mark		2		2	2	
		(ii)	No credit for 'Resistance decreases' Current increases or total circuit resistance decreases or any relevant resistance ratio (e.g. $\frac{V_{200}}{V_{\text{LDR}}}$) stated to increase or decrease as appropriate to the ratio (1) V_{200} goes up, but award mark only if argument presented so that this conclusion follows (1) no ecf from (c)(i)		2		2		
			Question 5 total	3	11	0	14	9	0